

DIMENSION THEORY & HYPERPLANE

$x + 5y = 3$  (straight line)

$y = mx + c$

← slope

$x = a$

A vertical line (D2)

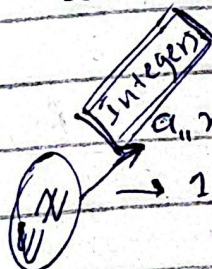
A point on  $\mathbb{R} = I(-\infty, \infty)$  (D1)

A plane in (3D)

Hyperplane in (4D)

" " nD

$ax + by = c \rightarrow$  No. unknowns define its geometry (2D Hyperplane)



$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b$  Hyperplane with D-n

1 value should be non-zero

Cartesian coordinate system

$x + 4y = 7$

$2x + 2y = 4$

$\mathbb{R}_2 = \mathbb{R} \times \mathbb{R}$

$\mathbb{R}_3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$

$\mathbb{R}_n = \mathbb{R} \times \dots \times \mathbb{R}$  (n-times)

If  $b=0$   
then it will be  
Homogeneous  
Eq.

If constants are zero

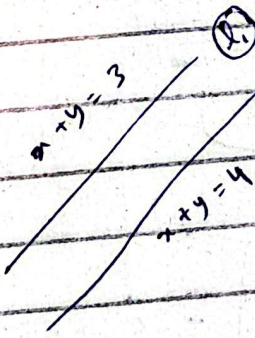
$a_{11}x + a_{12}y = b_1$

$a_{21}x + a_{22}y = b_2$

$l_1$   
 $l_2$  [2D]

Case

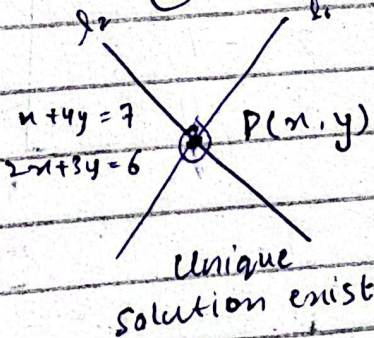
(I)



(No Solution)

Case

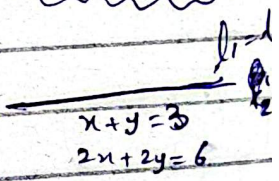
(II)



Unique solution exist

Case (III)

Overlapping



Multiple/Infinit solution exist

System of linear Equations {for multiple linear lines}

# SYSTEMS OF EQUATIONS

Case III

$$x + y + z = 3$$

$$(x2) \quad 2x + 2y + 2z = 6$$

$$(x3) \quad 3x + 3y + 3z = 9$$

$$x = t$$

$$\Rightarrow y + z = 3t$$

$$y = r$$

$$z = 3 - t - r$$

$$\begin{cases} (4) \quad x + y + 2z = 9 & \text{--- (1) } P_1 \\ (5) \quad 2x + 4y + 3z = 1 & \text{--- (2) } P_2 \\ (6) \quad 3x + 6y - 5z = 0 & \text{--- (3) } P_3 \end{cases}$$

→ when solution exist then

it will be consistent

otherwise it will be inconsistent.

→ Collection of arrays (numbers in order) [MATRIX]

Augmented matrix  $[A|B] =$

|   |   |    |   |
|---|---|----|---|
| 1 | 1 | 2  | 9 |
| 2 | 4 | -3 | 1 |
| 3 | 6 | -5 | 0 |

Row operations:  $-2R_1 + R_2$ ,  $-3R_1 + R_3$

Coefficient matrix

$$A X = B$$

0 (Homogeneous)

$\neq 0$  (Non-Homogeneous)

Elementary Row operation

→ This operation will not alter the solutions of systems.

→ Avoid Fraction  
→ If fractions then leave it as it is

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix} \quad \frac{1}{2} R_2$$

leading 1

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{bmatrix} \quad -3R_2 + R_3$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & -1/2 & -3/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$x + y + 2z = 9 \Rightarrow x = 1$   
 $y - \frac{7}{2}z = -\frac{17}{2} \Rightarrow y = 2$   
 $z = 3$

→ Gauss Elimination Method.  
Row Echelon Form

(Does not  
unique)

$X = (x, y, z) = (1, 2, 3)$  : comma delimited  
notation.

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Matrix  
Column  
Notation.

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \begin{array}{l} \swarrow \frac{1}{2} R_3 + R_2 \\ \searrow -2R_3 + R_1 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right] \searrow -R_2 + R_3$$

$$\sim \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Reduced Row Echelon Form.

→ Gauss Jordan Elimination Method (UNIQUE)

-Elementary Row operation does not change the solution.

$$AX=0 \text{ \& } AX=B$$

[A]  $[A|B|0]$

$$\begin{array}{cccccc|cc} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & & \\ \textcircled{1} & 3 & -2 & 0 & 2 & 0 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 & 0 \end{array}$$

$\rightarrow 4$  hyperplanes (row-wise)  
 $\rightarrow 6D$  (column-wise)  $\textcircled{4}$   
 $\mathbb{R}^6$  we will consider greater one (6)  
 $x \in \mathbb{R}^6$

$\begin{array}{l} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{array}$

matrix column rotation

$-R_1 + R_2$   
 $-2R_1 + R_4$

$$\sim \begin{array}{cccccc|cc} & 3 & -2 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & -2 & 0 & -3 & -1 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 & 0 \\ 0 & 0 & 4 & 8 & 0 & 18 & 6 & 0 \end{array}$$

$-R_2$   
 $-5R_2 + R_3$   
 $-4R_2 + R_4$

$$\sim \begin{array}{cccccc|cc} \textcircled{1} & 3 & -2 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \end{array}$$

$6x_6 = 2$   
 $x_6 = 1/3$

Quasi Elimination

$$\sim \begin{array}{cccccc|cc} \textcircled{1} & 3 & -2 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{1} & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{1} & 1/2 & 0 \end{array}$$

$-2R_3 + R_2$   
 $R_4 / 6$

Quasi Jordan:

$$\sim \begin{array}{cccccc|cc} 1 & 3 & -2 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$2R_2 + R_1$

$$\sim \begin{array}{cccccc|cc} 1 & 3 & 0 & 4 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$x_1 + 3x_2 + 4x_4 + 2x_5 = 0$   
 $x_3 + 2x_4 = 0$   
 $x_6 = 1/3$

Let  $x_4 = s$  put in eq (2)

$$x_3 + 2s = 0$$

$$x_3 = -2s$$

Put in eq (1)

$$x_1 + 3x_2 + 4s + 2x_5 = 0$$

$$x_1 + 3x_2 + 2x_5 = -4s$$

Let  $x_2 = r$   $x_5 = t$

$$x_1 + 3r + 2t = -4s$$

$$x_1 = -4s - 2t - 3r$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -4s - 2t - 3r \\ r \\ -2s \\ s \\ t \\ 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} -4 \\ 0 \\ 5 \\ 1 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} -2 \\ 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} t + \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1/3 \end{bmatrix} r$$

$$AX = 0$$

$$= r \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -4 \\ 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1/3 \end{bmatrix}$$

Theorem of

$$AX = B$$

→ Eq ↓ Unknown ↑ (Homo) Infinite Solutions.

# INCONSISTENT & CONSISTENT SYSTEMS

## TRACE, TRANSPOSE, MATRIX PRODUCTS AS LINEAR COMBINATIONS

— First find non-homogeneous then extract homogeneous one.

plane with 3D.

$$[A|B|0] = \left[ \begin{array}{ccc|c|c} 1 & 2 & -3 & 4 & 0 \\ 3 & -1 & 5 & 2 & 0 \\ 4 & 1 & a^2-14 & a+2 & 0 \end{array} \right] \begin{array}{l} -3R_1 + R_2 \\ -4R_1 + R_3 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c|c} 1 & 2 & -3 & 4 & 0 \\ 0 & -7 & 14 & -10 & 0 \\ 0 & -7 & a^2-2 & a-4 & 0 \end{array} \right] \begin{array}{l} R_2 \\ 7 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c|c} 1 & 2 & -3 & 4 & 0 \\ 0 & 1 & -2 & 10/7 & 0 \\ 0 & -7 & a^2-2 & a-4 & 0 \end{array} \right] \begin{array}{l} 7R_2 + R_3 \end{array}$$

$$\sim \left[ \begin{array}{ccc|c|c} 1 & 2 & -3 & 4 & 0 \\ 0 & 1 & -2 & 10/7 & 0 \\ 0 & 0 & a^2-16 & a-4 & 0 \end{array} \right]$$

$$(a^2-16)z = a-4$$

$$z = \frac{a-4}{a^2-16} \Rightarrow a \neq \pm 4$$

$$z = \frac{a-4}{(a+4)(a-4)}$$

$$z = \frac{1}{a+4} \Rightarrow a \neq -4$$

REMOVABLE DISCONTINUITY

① If  $a = -4$  (system has no solution)

②  $a \in \mathbb{R} \setminus \{-4\}$  (infinite many solutions)  $a \in (-\infty, -4) \cup (-4, \infty)$



$$\text{let } a=1 \Rightarrow \boxed{z = 1/5}$$

$$y - 2z = 10/7$$

$$y = \frac{2}{5} + \frac{10}{7} = \begin{pmatrix} 64 \\ 35 \end{pmatrix} \text{ (y)}$$

$$x = -2y + 3z$$

$$x = -2 \left( \frac{64}{35} \right) + 3 \left( \frac{1}{5} \right)$$

→ For each value of  $a$ , the system will give us unique solution.

Imp  
\*

$$AX = B$$

$$\begin{matrix} A & X & = & B \\ m \times n & n \times 1 & & m \times 1 \end{matrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$AX = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 \end{bmatrix} = x_1 \begin{matrix} \overset{C_1}{\begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix}} \end{matrix} + x_2 \begin{matrix} \overset{C_2}{\begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \end{bmatrix}} \end{matrix} + x_3 \begin{matrix} \overset{C_3}{\begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix}} \end{matrix}$$

$$= \text{Column}$$

$$= x_1 C_1 + x_2 C_2 + x_3 C_3$$

whenever you have a consistent system and has unique solution then solution is Linear combination.

$$[Q] \begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & 2 \end{bmatrix}, X = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

DECOMPOSE IN COLUMNS:

Product of Matrices

$$AX = 2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

LINEAR COMBINATION

In many cases, we decompose product of matrices in linear combinations of rows & columns.

$$\begin{bmatrix} 1 & -1 & -3 \end{bmatrix}_{1 \times 3} \begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} -16 & -18 & 25 \end{bmatrix}_{1 \times 3}$$

DECOMPOSE INTO ROWS:

$$1 \begin{bmatrix} -1 & 3 & 2 \end{bmatrix} + (-1) \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} + (-3) \begin{bmatrix} 2 & 1 & -2 \end{bmatrix} = \begin{bmatrix} -16 & -18 & 25 \end{bmatrix}$$

TRANSFORMATION

$$[Q] A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}_{2 \times 2}, X = \begin{bmatrix} a \\ b \end{bmatrix}_{2 \times 1}, AX = \begin{bmatrix} a \\ -b \end{bmatrix}_{2 \times 1}$$

$\mathbb{R}^2 \rightarrow \mathbb{R}^2$

Reflection about x-axis

TRANSPOSE: Rows into Columns 'or'  
Columns into Rows.

TRACE MATRIX ← ONLY FOR SQ. MATRIX  
DIAGONAL MATRIX

→ PROJECTION OF  $R_3$  onto  $R_2$  then  
 $R_3$  effects become 0 (coin by compressing)

\*  $AB \neq BA$

Closure  
Associative  
Identity  
Inverse  
Commutative  
Group

① if  $A = B$

$AB = BA$

$AA = AA$

$A^2 = A^2$  (Trivial Property)

$A/B = I$

② if  $AI = IA$

Recall

$AB = BA$

③  $A/B = 0 \Rightarrow AB = BA$

④  $A = B^{-1} \Rightarrow AB = BA$   
SQ. MATRIX

$AB = B^{-1}B = I$

$BA = BB^{-1} = I$

Determinant is always for sq. matrix.

Inverse

$\text{Det} \neq 0$  (Non-Singular / Invertible)  $\Rightarrow$  Then

$$\begin{aligned} ax+by &= k \\ cx+dy &= l \end{aligned} \longrightarrow \begin{aligned} a, b, c, d &\in \mathbb{Z} \\ AX &= B \end{aligned} \quad \begin{aligned} \text{inverse is} \\ \text{also invertible} \end{aligned}$$

$$[A|B|0] = \left[ \begin{array}{cc|c|c} a & b & k & 0 \\ c & d & l & 0 \end{array} \right] \xrightarrow[\substack{1/a R_1 \\ a \neq 0}]{} \begin{aligned} & \\ & \end{aligned}$$

$$\rightsquigarrow \left[ \begin{array}{cc|c|c} 1 & b/a & k/a & 0 \\ c & d & l & 0 \end{array} \right] \xrightarrow{-cR_1 + R_2}$$

$$\rightsquigarrow \left[ \begin{array}{cc|c|c} 1 & b/a & k/a & 0 \\ 0 & -\frac{bc}{a} + d & -\frac{kc}{a} + l & 0 \end{array} \right] \xrightarrow[\substack{a \\ ad-bc}]{} R_2$$

$$\rightsquigarrow \left[ \begin{array}{cc|c|c} 1 & b/a & k/a & 0 \\ 0 & 1 & \left(\frac{la-kc}{a}\right)\left(\frac{a}{ad-bc}\right) & 0 \end{array} \right] \xrightarrow[\substack{ad-bc \neq 0 \\ -b R_2 - R_1 \\ a}]{} \begin{aligned} & \\ & \end{aligned}$$

$$\rightsquigarrow \left[ \begin{array}{cc|c|c} 1 & 0 & \frac{k}{a} - \frac{b}{a} \left( \frac{la-kc}{ad-bc} \right) & 0 \\ 0 & 1 & \frac{la-kc}{ad-bc} & 0 \end{array} \right] \xrightarrow{\left( \frac{la-kc}{ad-bc} \right) \left( \frac{1}{a} \right) + \frac{k}{a}}$$

$$y = \frac{la-kc}{ad-bc} \quad ad-bc \neq 0$$

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$: A_{n \times n} \longrightarrow \mathbb{R}$$

defined as

$$|B_{\text{sys}}| = \alpha \in \mathbb{R}$$

# ⇒ ALGEBRAIC PROPERTIES OF MATRICES (EX 14)

$$\star A^{-1} = \frac{1}{|A|} \quad \star (A^{-1})^t = (A^t)^{-1} \quad \star (AB)^{-1} = B^{-1} A^{-1}$$

$$\star (AB)^t = B^t A^t \quad \star (A \pm B)^t = A^t \pm B^t \quad \star (A^{-1})^{-1} = A$$

## Definition; ELEMENTARY MATRICES

A matrix is said to be elementary if it is obtained by performing A SINGLE ELEMENTARY ROW OPERATION ON IDENTITY MATRIX. (SEE LAST PAGE)

→ coefficient matrix

$$[A | I_{2 \times 2}] = \left[ \begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ -1 & 3 & 0 & 1 \end{array} \right] \quad \text{Augmented}$$

$$[A | I_{2 \times 2}] = \left[ \begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ -1 & 3 & 0 & 1 \end{array} \right] \xrightarrow{R_1 + R_2, \text{ (1)}} E_1 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad E_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sim \left[ \begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array} \right] \xrightarrow{2R_2 + R_1, \text{ (2)}} E_2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\sim \left[ \begin{array}{cc|cc} 1 & 0 & 3 & 2 \\ 0 & 1 & 1 & 1 \end{array} \right] \quad A^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$E_2 E_1 A = I_{2 \times 2}$$

$$E_2^{-1} (E_2 E_1 A) = E_2^{-1} I_{2 \times 2}$$

$$E_1 A = E_2^{-1} I_{2 \times 2}$$

$$E_1^{-1} (E_1 A) = E_1^{-1} (E_2^{-1} I_{2 \times 2}) = E_1^{-1} (E_2^{-1} I)$$

$$A = E_1^{-1} E_2^{-1} I$$

$$A^{-1} = (E_1^{-1} E_2^{-1} I)^{-1} \Rightarrow A^{-1} = E_1 E_2 I_{2 \times 2}$$

\*  $\det \neq 0$   
for finding Inverse

$AX=B$

Q)  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$  Plane (3-D)  
3x3

$$[A | I_{3 \times 3}] = \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \textcircled{1} \\ -2R_1 + R_2 \\ -R_1 + R_3 \end{array}$$

$$\sim \left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right]$$

$K=7$

$E_k E_{k-1} \dots E_2 E_1 A = I_{n \times n}$

$$\sim \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 10 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \rightarrow A^{-1}$$

\* These all matrices and inverse of these matrices are invertible  
 $E_7 E_6 E_5 \dots E_3 E_2 E_1 A = I_{3 \times 3}$

$E_7^{-1} (E_7 E_6 E_5 \dots E_2 E_1 A) = E_7^{-1} I_{3 \times 3}$

\*\*  $A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_7^{-1} I_{3 \times 3}$

Decomposition.

\*\*\*  $A^{-1} = E_7 E_6 E_5 \dots E_1 A I_{3 \times 3}$

$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$   $\textcircled{1}$

$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$   $\textcircled{2}$

$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 2 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$   $E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & | & 1 & 0 & 0 \\ -2 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$   $\textcircled{2} 2R_1 + R_2$

## → SQUARE + INVERTIBLE MATRICES

$$AX = B$$

$$A^{-1}(AX) = A^{-1}B$$

$$X = A^{-1}B$$

Let  $B_1 = \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix}$ ,  $B_2 = \begin{bmatrix} 1 \\ 6 \\ -6 \end{bmatrix}$ ,  $B_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ,  $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ ,  $b_i \in \mathbb{Z}$

$$X = A^{-1}B_1$$

$$= \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix} = \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix}$$

$3 \times 3 \quad 3 \times 1 \quad 3 \times 1$

$$X = A^{-1}B_3$$

$$= \begin{bmatrix} 11 \\ 11 \\ 11 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$3 \times 3$

$$x = y = z = 0$$

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 1 & 0 & 8 & b_3 \end{array} \right]$$

Reduced  
echelon  
forms

always  
Identity  
when  
matrix  $A$  is  
invertible

$$\sim \left[ \begin{array}{ccc|c} 1 & 0 & 0 & -40b_1 + 16b_2 + 9b_3 \\ 0 & 1 & 0 & 13b_1 - 5b_2 - 3b_3 \\ 0 & 0 & 1 & 5b_1 - 3b_2 - b_3 \end{array} \right]$$

trivial  
solution

→ The system will  
always be consistent  
for every  $b_1, b_2, b_3 \in \mathbb{Z}$   
→ For each fix  $b_1, b_2, b_3$   
the system will give  
you unique solution.

$$x_1 = -40b_1 + 16b_2 + 9b_3$$

$$x_1 + x_2 + 2x_3 = b_1$$

$$x_1 + x_3 = b_2$$

$$2x_1 + x_2 + 3x_3 = b_3$$

$$[A|B] = \left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{array} \right]$$

$$\det(A) = 0$$

→ 1 or more than 1  
row will be of zeros

$$\sim \left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array} \right]$$

① if  $0x_1 + 0x_2 + 0x_3 \neq 0$   
inconsistent.

② If  $b_3 - b_2 - b_1 = 0 \Rightarrow b_3 = b_2 + b_1$   
Consistent

The system is consistent for every  $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$   
 Non-homogeneous  $\rightarrow$  No. of eq. are 2 and unknown = 3.  
 $\rightarrow$  Infinite Solutions.  $= 0$   
 $\rightarrow$  No Solution  $\neq 0$

$AX = B$   
 $A = LU$   
 $LUX = B$   
 $LY = B$   
 $LY = B$

$U = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \end{bmatrix}$  Row Echelon Form.  
 $L = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$   $i < j$   
 $i = j$

A square matrices is said to be diagonal if all the values above and below from principle diagonal. non-sq.  $\neq$  diag.

$D^k = \begin{bmatrix} d_{11}^k & 0 & 0 \\ 0 & d_{22}^k & 0 \\ 0 & 0 & d_{33}^k \end{bmatrix}$   $k \geq 1$

$DD \checkmark$   
 $D \times D \times$

(Power always be positive)  $\Rightarrow$  if  $D^{-k}$   $\Rightarrow (D^{-1})^k$   
 Find Inverse first.  
 $(D^T)^s = D^{T \times s}$

Theorem

$\rightarrow$  Diagonal matrix exist only if principle diagonal all values are non-zero.

$\rightarrow$  Symmetric  $\pm$  Symmetric = Symmetric.

Scalar  $\times$  Symmetric = Symmetric

Transpose of " = "

load on on

$\iff A$  is invertible + sq.

\*  $(A^{-1})^t = (A^t)^{-1}$  For Invertible Matrices

\* \* \*  $(AB)^{-1} = B^{-1}A^{-1}$

\*  $(AB)^t = B^t A^t$

$\rightarrow$  PROPERTIES OF INVERSES & TRANSPOSE

$\rightarrow$  EQUIVALENCE THEOREM.

Q  $A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$

$AX = 0/B$

Sol

$A^{-1}$  exists.

$X^* = A^{-1}B/0$

$A(B) = I$

$(B)A = I \text{ Adj } A$

$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} ad-bc & -ab+ab \\ cd-cd & -bc+ad \end{bmatrix} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$

$= (ad-bc) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(2)  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(3)  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} k \\ l \end{bmatrix} \sim \begin{bmatrix} 1 & b/a \\ 0 & \frac{ad-bc}{a} \end{bmatrix} = \begin{bmatrix} - \\ - \end{bmatrix}$

$\det(A) = ad-bc \neq 0$

$\begin{pmatrix} | & | \\ & \\ f & x & y \end{pmatrix} = A_{n \times n} \rightarrow \mathbb{R}$   
 $\rightarrow |A| = \alpha \in \mathbb{R}$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

\* Minor of  $a_{ij} = m_{ij}$   
 $a_{11} \rightarrow m_{11} =$  We will drop row & column of this entry then write sub-matrix (remaining)

$$\det(d) = d$$

$$a_{21} = m_{21} = b$$

$$a_{12} = m_{12} = c$$

$$a_{22} = m_{22} = a$$

\* Cofactors

$$C_{ij} = (-1)^{i+j} m_{ij}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$C_{ij} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$$(C_{ij})^T = \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix}$$

$$(C_{ij})^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad \text{Adj } A$$

$$* A \cdot \text{Adj}(A) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}$$

$$\begin{aligned} C_{11} &= 12 \\ C_{21} &= 4 \\ C_{31} &= 12 \\ C_{12} &= -6 \\ C_{22} &= 2 \\ C_{32} &= -16 \\ C_{13} &= -16 \\ C_{23} &= 16 \\ C_{33} &= 16 \end{aligned}$$

$$= \begin{bmatrix} a_{11}C_{11} + a_{12}C_{21} + a_{13}C_{31} & a_{11}C_{12} + a_{12}C_{22} + a_{13}C_{32} & a_{11}C_{13} + a_{12}C_{23} + a_{13}C_{33} \\ a_{21}C_{11} + a_{22}C_{21} + a_{23}C_{31} & a_{21}C_{12} + a_{22}C_{22} + a_{23}C_{32} & a_{21}C_{13} + a_{22}C_{23} + a_{23}C_{33} \\ a_{31}C_{11} + a_{32}C_{21} + a_{33}C_{31} & a_{31}C_{12} + a_{32}C_{22} + a_{33}C_{32} & a_{31}C_{13} + a_{32}C_{23} + a_{33}C_{33} \end{bmatrix}$$

$\det A$      $\det A$      $\det A$

$$\begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$

$$\det(A) = 64$$

same row same cofactor

$$* a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} = (4)(4) + 6(2) + 3(16) = 64$$

$$* a_{21}(C_{11}) + a_{22}(C_{12}) + a_{23}(C_{13}) = (1)(12) + 6(6) + 3(-16) = 0$$

$$* a_{13}C_{13} + C_{23}C_{23} + a_{33}C_{33} = -1(16) + 3(16) + (0)(16) = 64$$

Row Column = det

row · column = 0.

2nd Column

diff. Column.

$$* a_{13}C_{12} + a_{23}C_{22} + a_{33}C_{32} = -1(6) + 2(3) + 0(-19) = 0$$

$$A \text{ adj } A = \begin{bmatrix} \det A & 0 & 0 \\ 0 & \det A & 0 \\ 0 & 0 & \det A \end{bmatrix}$$

$$= \det A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$* A \text{ adj } A = \det(A) I$$

$$A^{-1} = \frac{\text{adj } A}{\det A}$$

$$\det A = ad - bc$$

$$a_{11}C_{11} - a_{12}C_{12}$$

$$* \text{Adj}(A) = \det(A) A^{-1}$$

$$* X = A^{-1} B \text{ or } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{\text{adj}(B)}{\det(A)}$$

$$B = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{33} & C_{33} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$= \frac{1}{\det A} \begin{bmatrix} b_1 C_{11} + b_2 C_{22} + b_3 C_{33} \\ b_1 C_{12} + b_2 C_{23} + b_3 C_{32} \\ b_1 C_{13} + b_2 C_{23} + b_3 C_{33} \end{bmatrix}$$

CRAMER'S RULE

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{b_1 C_{11} + b_2 C_{22} + b_3 C_{33}}{\det(A)}$$

$$\begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix} = A$$

$$x_2 = \frac{\det(A_2)}{\det(A)} = \frac{b_1 C_{12} + b_2 C_{23} + b_3 C_{32}}{\det(A)}$$

$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{b_1 C_{13} + b_2 C_{23} + b_3 C_{33}}{\det(A)}$$

# DETERMINANTS & ITS PROPERTIES

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{bmatrix}$$

determinant scalar

$$\det(A) = |A| = \begin{vmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{vmatrix}$$

② → Common from how many number of rows.

$$\begin{vmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 0 & 1 & 2 \end{vmatrix} = (-1) \begin{vmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 0 & 1 & 2 \end{vmatrix}$$

③ → Common from 1 Row

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{vmatrix} \xrightarrow{(-1)(3)} \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 6 & 1 \end{vmatrix}$$

## PROPERTIES

- ①  $R_2 \leftrightarrow R_1$   
 $\det A = -\det B$
- ②  $\det(kA) = k^n \det A$

$$\xrightarrow{(-1)(3)} \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{vmatrix}$$

$-2R_1 + R_3$

$-10R_2 + R_3$

③

triangular matrix  
det = product of diagonal entries.

$$\xrightarrow{(-1)(3)} \begin{vmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{vmatrix}$$

$$\xrightarrow{(-1)(3)(1)(1)(-55)} = 1 \cdot 65$$

## PROPERTIES

Another matrix  
(order does not matter)

- ③  $\det(EA) = \det E \det A$
- ④  $\det(AB) = \det A \det B \rightarrow$  order must be same.
- ⑤  $\det(A+B) \neq \det A + \det B$
- ⑥  $\det(A^{-1}) = \frac{1}{\det(A)}$   
 $A^{-1}A = I$   
 $\det(A^{-1}A) = \det(I)$   
 $\det(A^{-1}A) = 1$   
 $\det A^{-1} = \frac{1}{\det A}$

→ order will be same

$$\boxed{\det A^{-1} = \frac{1}{\det A}}$$

# EIGEN VALUES &

# EIGEN VECTORS

$$A X = \lambda X$$

$X = ?$  (EIGEN VECTOR)

$\lambda = ?$  (Eigen Value)

Q: For what value of  $\lambda$  (lambda) the given system will be consistent?

$Q =$

$$x_1 + 3x_2 = \lambda x_1 + 0x_2$$

$$4x_1 + 2x_2 = \lambda x_1 + 0x_2$$

Matrix notation

$$\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\lambda I$

$A$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \lambda - 1 & -3 \\ -4 & \lambda - 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- (1)}$$

$$(\lambda I - A) X = 0 \quad \text{--- (2)}$$

For homogeneous (constants = 0)

$$A^* X = 0 \quad \text{--- (3)}$$

For non-trivial solution (neg of Equivalence Theorem)  
 → Matrix will not be invertible and det will be zero.

$$\det(A^*) = 0$$

or

$$\det(\lambda I - A) = 0$$

$$\det \begin{vmatrix} \lambda - 1 & -3 \\ -4 & \lambda - 2 \end{vmatrix} = 0$$

$$(\lambda - 1)(\lambda - 2) - 12 = 0$$

$$\lambda = -2, 5$$

$$\lambda_1 = -2$$

$$\lambda_2 = 5$$

EIGEN VALUES

characteristic eq.  
 Auxiliary eq.

Complex / Irrational roots  $\rightarrow$  answer will be in pair.

Case I:  $\lambda_1 = -2$ ,  $X = ?$

we will use eq ①

$$\begin{bmatrix} -2 & -1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & -3 \\ -4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{array}{l|l} -3x_1 - 3x_2 = 0 & x_1 - x_2 = 0 \\ -4x_1 - 4x_2 = 0 & x_1 - x_2 = 0 \end{array}$$

$$x_1 + x_2 = 0 \quad n = 2$$

$$x_2 = t \quad r = 1$$

$$x_1 = -t \quad n - r = 1$$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix} \rightarrow \text{EIGEN VALUES}$$

$t \in \mathbb{R}$  (Infinite) solutions.

Case - II:  $\lambda_2 = 5$ ,  $X = 5 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$A = \begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix}$$

$$3x_1 + 2x_2 = \lambda x_1 + 0x_2$$

$$x_1 + 2x_2 = \lambda x_2 + 0x_1$$

$$\begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \lambda - 3 & -2 \\ -1 & \lambda + 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \text{①}$$

$$(\lambda I - A) X = 0 \rightarrow \text{②}$$

$$A^* X = 0 \rightarrow \text{③}$$

$$\det(A^*) = 0$$

$$\det (\lambda I - A) = 0$$

$$\det \begin{vmatrix} \lambda - 3 & -2 \\ -1 & \lambda + 2 \end{vmatrix} = 0$$

$$(\lambda - 3)(\lambda + 2) - 2 = 0$$

$$(\lambda - 3)(\lambda + 2) = 2$$

$$\lambda^2 - 3\lambda + 2\lambda - 6 = 2$$

$$\lambda^2 - \lambda = 8$$

$$\lambda^2 - \lambda - 8 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Always remain  
1 equation.

$$\lambda = \frac{-1 \pm \sqrt{1 - 4(-8)}}{2(1)}$$

$$\lambda = \frac{-1 + \sqrt{33}}{2}, \quad \lambda = \frac{-1 - \sqrt{33}}{2}$$

~~ack~~

Infinite  
many solutions

$$\lambda_{1,2} = \frac{-1 \pm \sqrt{33}}{2}$$

$$\frac{-1 + \sqrt{33}}{2}, \quad \frac{-1 - \sqrt{33}}{2}$$

$$\begin{bmatrix} \frac{-1 + \sqrt{33}}{2} & -2 \\ -1 & \frac{-1 + \sqrt{33}}{2} + 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{-1 + \sqrt{33}}{2} x_1 - 2x_2 = 0$$

$$-x_1 + \frac{-1 + \sqrt{33}}{2} x_2 = 0$$

$$n = 2$$

$$r = 1$$

$$n - r = 1$$

Dot product for  $\mathbb{R}_2$  &  $\mathbb{R}_3$   
cross product valid for  $\mathbb{R}-3$

$\mathbb{R}_4$  & above  $\rightarrow$  inner product

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$X = (x_1, x_2) = t(-1, 1)$$

$$x_1 = -t$$

$$x_2 = t$$

$$x = 0 - t$$

$$y = 0 + t$$

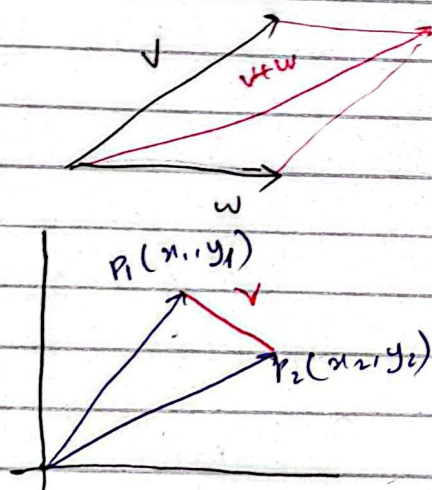
Parametric

Eq. of straight line

$$x = x_0 + t x_1$$

$$y = y_0 + t x_2$$

## Euclidean Vector Space



euclidean  
 $n$ -space =  $\mathbb{R}^n$

Let  $v \in \mathbb{R}^n$

$\Rightarrow n$ -components

$$v = [v_1, \dots, v_n] \in \mathbb{R}^n$$

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\vec{P_1 P_2} = \vec{OP_2} - \vec{OP_1} = (x_2 - x_1, y_2 - y_1)$$

$\rightarrow$  Sum of Vectors Properties (From Book)

$$0 \leq n$$

Let  $w, v_i \in \mathbb{R}^n$

\* Parallel vectors are scalar multiple of each other.

Vectorial Form  $w = (w_1, w_2, w_3)$

$$= w_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + w_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + w_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= w_1 \hat{i} + w_2 \hat{j} + w_3 \hat{k}$$

Let  $v_1, v_2, v_3, w \in \mathbb{R}^3$   
 $w$  is a linear combination of  $v_1, v_2, v_3$

$$w = k_1 v_1 + k_2 v_2 + k_3 v_3$$

→ If matrix is invertible then answer will be unique according to Equivalence theorem ( $\det \neq 0$ )

→ But if ( $\det = 0$ )  $\Rightarrow$  one row will be ~~non~~ zeros and infinite many solutions.  $\rightarrow$  depends on R.H.S coefficients.

→ negation

→ we can use magnitude/length for vectors. we use the term "norm". Norm is a function and a vector  $V$  that has real values.

$$\|u-v\|^2 + \|u+v\|^2 = 2 \{ \|u\|^2 + \|v\|^2 \}$$

Parallelogram Equation/ for vector Law

→ euclidean inner product / dot product / scalar product.

See Pic

→  $R_n$  or above inner product.

→ orthogonal used for matrices instead of perpendicular.

A vector:

→ Each row of  $A$  produce 0 by orthogonality.

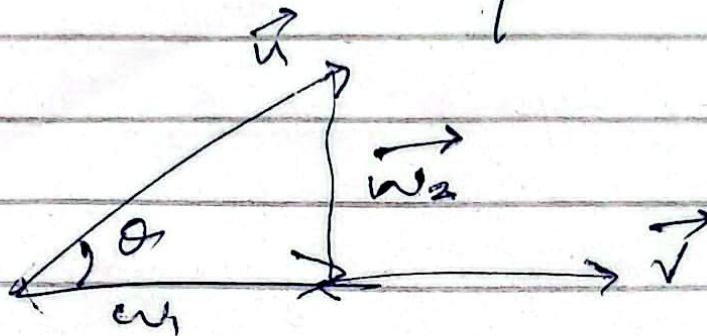
→ null space  $\rightarrow$  null vectors always orthogonal.

→  $y_{mc}$

marginal  
Slope.

Vector

Projection : Vector image / Shadow.



vector can be :

- ① matrix
- ② function
- ③ Infinite sequence

$$+(u_1 - v_1)^2 + \dots + (u_n - v_n)^2$$

$$\left| \frac{u \cdot v}{\|u\| \|v\|} \right| \leq 1 \quad \checkmark$$

by Schwarz inequality

$$\sqrt{u_1^2 + \dots + u_n^2} \sqrt{v_1^2 + \dots + v_n^2}$$

\* Law of Cosines

$$\|\vec{P_1 P_2}\|^2 = \|u\|^2 + \|v\|^2 - 2|u||v|\cos \theta$$

$$\vec{P_1 P_2} = v - u$$

$$u, v \in \mathbb{R}^3$$

$$2(|u||v|\cos \theta) = \|u\|^2 + \|v\|^2 - \|v - u\|^2$$

$$u \cdot v = \frac{1}{2} (\|u\|^2 + \|v\|^2 - \|v - u\|^2) = |u||v|\cos \theta$$

$$\times \quad \|u - v\|^2 + \|u + v\|^2 = 2\{\|u\|^2 + \|v\|^2\} \quad u, v \in \mathbb{R}^n$$

$$\times \quad \|u - v\|^2 - \|u + v\|^2 = 4(u \cdot v)$$

$$\checkmark \quad \|v - u\|^2 = (v - u) \cdot (v - u) = v \cdot v - v \cdot u - u \cdot v + u \cdot u = \|v\|^2 - 2(u \cdot v) + \|u\|^2 \rightarrow (1)$$

$$\times \quad \|v + u\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v) \rightarrow (2)$$

$$\left. \begin{aligned} & u_1 v_1 + u_2 v_2 + \dots + u_n v_n \in \mathbb{R} \end{aligned} \right\}$$

$\times$  if  $u = \|u\|^2$   
 $\times$   $\|u\| = \sqrt{u \cdot u}$   
 $\times$   $u \in \mathbb{R}^n$   
 $\times$   $u \in \mathbb{R}^n$   
 $\times$   $u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$   
 $\times$   $\frac{1}{\|u\|} \frac{1}{\|v\|}$   
 $\times$   $\frac{1}{\|u\|} \frac{1}{\|v\|}$

$$\|u+v\|^2 =$$

$$d = \|\vec{p}_1 \vec{p}_2\|$$

$$u, v \in \mathbb{R}^3$$

$$\|u-v\| = \sqrt{(u_1-v_1)^2 + (u_2-v_2)^2 + \dots + (u_n-v_n)^2}$$

$$\left| \frac{u \cdot v}{\|u\| \|v\|} \right| \leq 1$$

$$-1 \leq \frac{u \cdot v}{\|u\| \|v\|} \leq 1 \quad (\text{Cauchy-Schwarz inequality})$$

$$|u \cdot v| \leq \|u\| \|v\|$$

$$|u_1 v_1 + \dots + u_n v_n| \leq \left( \sqrt{u_1^2 + \dots + u_n^2} \right) \left( \sqrt{v_1^2 + \dots + v_n^2} \right)$$

$\times$  Law of Cosines

$$\|\vec{p}_1 \vec{p}_2\|^2 = \|u\|^2 + \|v\|^2 - 2|u||v|\cos \theta$$

$$\vec{p}_1 \vec{p}_2 = v - u$$

$$u, v \in \mathbb{R}^3$$

$$2(|u||v|\cos \theta) = \|u\|^2 + \|v\|^2 - \|v-u\|^2$$

$$u \cdot v = \frac{1}{2} \left( \|u\|^2 + \|v\|^2 - \|v-u\|^2 \right) = u_1 v_1 + u_2 v_2 + \dots + u_n v_n \in \mathbb{R}$$

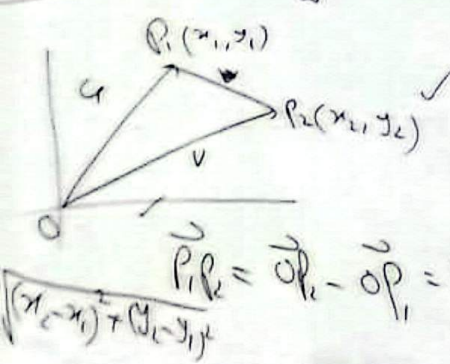
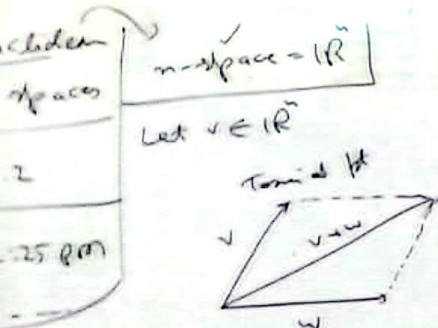
$$\times \|u-v\|^2 + \|u+v\|^2 = 2(\|u\|^2 + \|v\|^2) \quad u, v \in \mathbb{R}^n$$

$$\times \|u-v\|^2 - \|u+v\|^2 = -4(u \cdot v)$$

$$\checkmark \|v-u\|^2 = (v-u) \cdot (v-u) = v \cdot v - v \cdot u - u \cdot v + u \cdot u = \|v\|^2 - 2(u \cdot v) + \|u\|^2$$

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v) \quad \text{--- (2)}$$

$$u_1 v_1 + u_2 v_2 + \dots + u_n v_n \in \mathbb{R}$$



Define  $A$  s.t.  $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$  reflection about  $y$  axis.

$v = (v_1, \dots, v_n) \in \mathbb{R}^n$

$v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$

$w = (w_1, w_2, w_3) \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$   
 $= w_1(1, 0, 0) + w_2(0, 1, 0) + w_3(0, 0, 1)$

Let  $w_1, v_1, \dots, v_n \in \mathbb{R}^n$

$\mathbb{R}^n \subseteq \mathbb{R}^n$

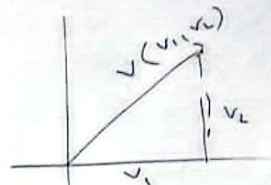
Let  $v_1, w, v_2 \in \mathbb{R}^3$

$w = w_1 \hat{i} + w_2 \hat{j} + w_3 \hat{k}$

$w = (w_1, w_2, w_3) = k_1(3, 2, 5) + k_2(2, 4, 6) + k_3(7, 8, 9)$

$w = k_1 v_1 + k_2 v_2 + k_3 v_3$

$\begin{cases} 3k_1 + 2k_2 + 7k_3 = w_1 \\ 2k_1 + 4k_2 + 8k_3 = w_2 \\ 5k_1 + 6k_2 + 9k_3 = w_3 \end{cases} \Rightarrow AX = B/w$



$u \cdot v = \|u\| \|v\| \cos \theta$   
 $u \cdot u = \|u\| \|u\| \cos(0)$   
 $\|u\|^2 + \|u\|^2 = \|u\|^2$

$\|v\| = \sqrt{v_1^2 + v_2^2}$

$\| \cdot \| : \text{Set of all vectors} \rightarrow \mathbb{R}$

$\|v\| = \alpha \in \mathbb{R}$

$u \cdot u = \|u\|^2$   
 $\|u\| = \sqrt{u \cdot u}$   
 $\|u\| = \sqrt{u_1^2 + u_2^2}$   
 $u = (u_1, u_2)$   
 If  $u \in \mathbb{R}^n$   
 $\|u\| = \sqrt{u_1^2 + \dots + u_n^2}$

$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3$

$\frac{\vec{v}}{\|\vec{v}\|}$   
 Unit vector

786

Let  $u, v \in \mathbb{R}^3$

\*  $u$  - column matrix  
 $v$  - " matrix

$$\underline{u \cdot v} = u^t v = v^t u$$

$$u = \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}, v = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$$

$3 \times 1 \quad \quad 3 \times 1$

$$t = \{1 \ -3 \ 5\} \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix}$$

$1 \times 3 \quad \quad 3 \times 1$

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3 = -7$$

n-space =  $\mathbb{R}^n$

Dot product &  
matrix products

\*  $u$  - row matrix  
 $v$  - column "

$$u \cdot v = u v = v^t u^t$$

$$u = \begin{bmatrix} 1 & -3 & 5 \end{bmatrix}$$

$$v = \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix}$$

$$u v = \begin{bmatrix} 1 & -3 & 5 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix} =$$

projection of  
 $\vec{u}$  onto  $\vec{v}$

\*  $u$  - column "  
 $v$  - row "

$$u \cdot v = v u = u^t v^t$$

Quiz @ 12.25 pm  
@ 5/6/

\*  $u$  &  $v$  - row matrices

$$u \cdot v = u v^t = v u^t$$

$$= 5 - (2 + 0) = \begin{bmatrix} -7 \end{bmatrix}_{1 \times 1} = -7$$

$$x \otimes y \quad x_{n \times n}, u, v \rightarrow n \times 1$$

$$\begin{aligned} & \boxed{A u \cdot v = u \cdot A^t v} \\ & \boxed{u \cdot A v = A^t u \cdot v} \end{aligned}$$

$$\begin{bmatrix} A \end{bmatrix}_{n \times n} \begin{bmatrix} u \end{bmatrix}_{n \times 1} = \begin{bmatrix} A u \end{bmatrix}_{n \times 1}$$

$$AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

$$a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = (a_{11}, a_{12}, a_{13}) \cdot (b_{11}, b_{21}, b_{31}) = r_1 \cdot c_1$$

$$AB = \begin{bmatrix} r_1 \cdot c_1 & r_1 \cdot c_2 & r_1 \cdot c_3 \\ r_2 \cdot c_1 & r_2 \cdot c_2 & r_2 \cdot c_3 \\ r_3 \cdot c_1 & r_3 \cdot c_2 & r_3 \cdot c_3 \end{bmatrix}$$

$$X = (x_1, x_2, x_3)$$

$$A \overset{\checkmark}{X} = B / 0 \overset{\checkmark}$$

$$AX = \begin{bmatrix} r_1 \cdot X \\ r_2 \cdot X \\ r_3 \cdot X \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left. \begin{aligned} r_1 \cdot X &= 0 \\ r_2 \cdot X &= 0 \\ r_3 \cdot X &= 0 \end{aligned} \right\} AX = 0$$

$$(a_{11}, a_{12}, a_{13}) \cdot (x_1, x_2, x_3) = 0$$

$$AX = 0 \quad X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{n \times 1}$$

$$X \neq 0$$

$$(a_1, \dots, a_n) \cdot (x_1, \dots, x_n)$$

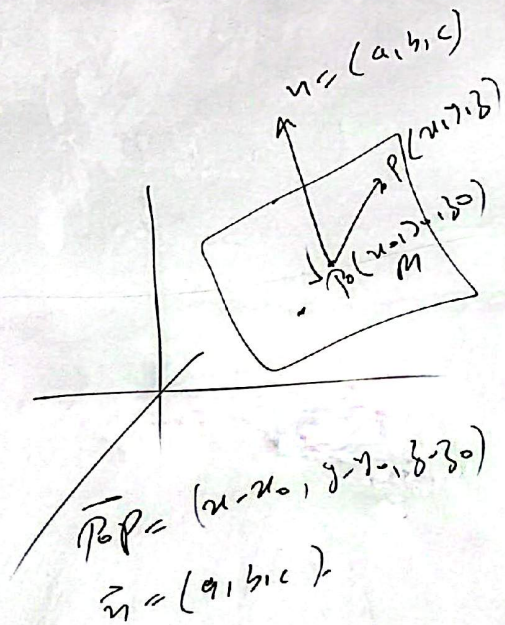
$$(z_1, b_1) = x_1 \cdot c_1$$

$$\begin{cases} 0 \\ 0 \\ 0 \end{cases} \} AX=0$$

$$(a_1, b_1) \cdot (x_1, y_1) = 0$$

$$X = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}_{n+1}$$

$$(a_1, \dots, a_n) \cdot (x_1, \dots, x_n) = 0$$



$$\vec{P_0P} \cdot \vec{n} = 0$$

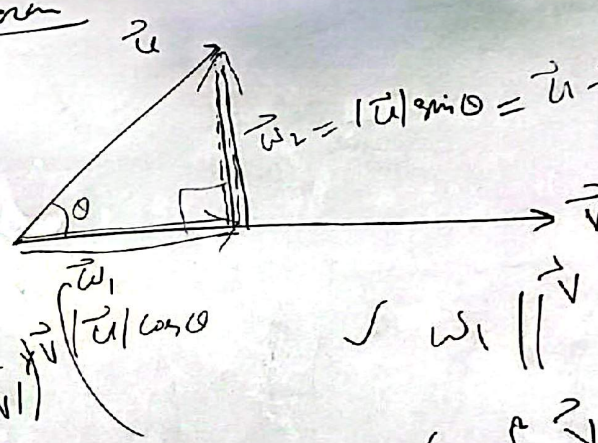
$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$ax + by + cz = d$$

projection theorem

$$\vec{w}_1 = \text{proj}_{\vec{v}} \vec{u}$$

$$= (|\vec{u}| \cos \theta) \left( \frac{\vec{v}}{|\vec{v}|} \right)$$



$$\vec{w} = \vec{w}_1 + \vec{w}_2$$

$$\vec{w}_2 = |\vec{u}| \sin \theta = \vec{u} - \vec{w}_1 = \vec{u} - (\text{proj}_{\vec{v}} \vec{u})$$

$$\sqrt{w_1} \parallel \vec{v}$$

$$\sqrt{w_2} \perp \vec{v}$$

$$\frac{\vec{v}}{|\vec{v}|} \times$$

$$\vec{w}_1 = \frac{(\vec{u} \cdot \vec{v})}{|\vec{v}|^2} \vec{v} = \left( \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v}$$

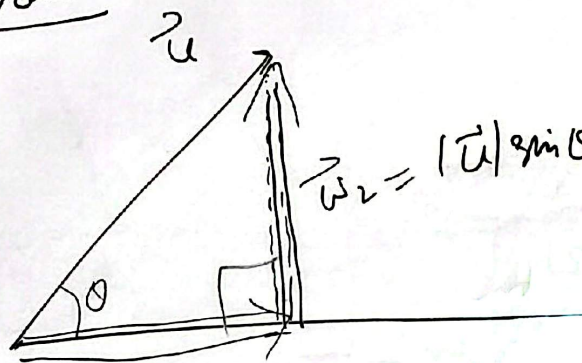
$$\vec{w}_2 = \text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}$$

projection theorem

$$\vec{w} = \vec{w}_1 + \vec{w}_2 \checkmark$$

$$\vec{w}_1 = \text{proj}_{\vec{v}} \vec{u}$$

$$= (|\vec{u}| \cos \theta) \frac{\vec{v}}{|\vec{v}|}$$



$$\vec{w}_2 = |\vec{u}| \sin \theta = \vec{u} - \vec{w}_1 = \vec{u} - (\text{proj}_{\vec{v}} \vec{u}) \checkmark$$

$$\vec{w}_1 \parallel \vec{v}$$

$$\frac{\vec{v}}{|\vec{v}|} \times$$

$$\vec{w}_1 = \frac{|\vec{u}| |\vec{v}| \cos \theta}{|\vec{v}| |\vec{v}|} \vec{v} = \left( \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v}$$

$$\vec{w}_1 = \text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v}$$

①

```
void func(int *p)
{
    p++;
    *p = *p + 10;
}
```

```
int main()
```

```
{
    int arr[] = {5, 10, 15};
```

```
    int *p = arr;
```

```
    func(p);
```

```
    cout << *p << endl;
```

```
    p--;
```

```
    cout << *p << endl;
```

```
}
```

②

```
void func(int *p)
```

# FREE VARIABLE THEOREM FOR Homogeneous System of Eqs (Theorem)

Whenever we have a system of eq with more unknowns than eq we will always get infinite many solutions.

(Note: From solution of non-homo, homo solution can be extracted)

$AX=0$  (solution just add constant vector)

$X = (x_1 + x_2 + x_3 + x_4)$

Eq = unknown [Identity matrix] [Diagonal]

If unknowns > eq → Infinite many solutions.

$AX=B$

if  $X=[0]$

$A[X]=B$

Let  $X$  be the solution

$A[0]=B$

$[0]=B$

Non-homogeneous is a trivial solution only if  $X=[0]$

If  $B$  is not zero then it will be non-homogeneous

$[0] \neq [B]$  (Inconsistent)

Conclusion, non-homogeneous always unique infinite many solutions and non-trivial

If  $A_{m \times n} \xrightarrow{ERO} NR$

Either it will have row of rows

OR it will be equal to  $I_{n \times n}$

(Trivial solution)

$\begin{bmatrix} 0 & 0 \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} x \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \end{bmatrix}$

Let  $r$  be the # of non-zero rows → # of free variables must be equal to non-zero rows.

If  $A_{m \times n} \xrightarrow{ERO} NR \rightarrow$  Above is valid here as well.

## ELEMENTARY MATRICES

$E_k E_{k-1} \dots E_2 E_1 A = I_n$  (\*)

$A = E_1^{-1} E_2^{-1} \dots E_{k-1}^{-1} E_k^{-1} I_n$  (\*\*)

$A^{-1} = E_k E_{k-1} \dots E_2 E_1 I_n$  (\*\*\*)

$(AB)^{-1} = B^{-1} A^{-1}$   
(Order change)

## EQUIVALENCE / NEGATION OF EQUIVALENCE THEOREM

we are given sq. matrix  $n \times n$

1.  $A$  is <sup>not</sup> invertible

2.  $AX=0$  (Homogeneous system) has only the trivial solution ( $X=0$ )

should be in order.

3. The reduced row echelon form of  $A$  is Identity.
4.  $A$  is expressible as product of elementary matrices.
5.  $AX=B$  (non homo) is consistent for every  $B = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1}$ .
6.  $AX=B$  has exactly 1 solution for every  $B = \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n \times 1}$  matrix.

→ At a time either all are true or all are false.  
→ System consistent (Identity) if it is not identity then 1 row = 0.

NEGATION:

~~NEGATION~~  $p \rightarrow q \equiv \neg q \rightarrow \neg p$

7.  $\det(A) \neq 0$  non zero when sq. matrix is invertible.
- trivial.